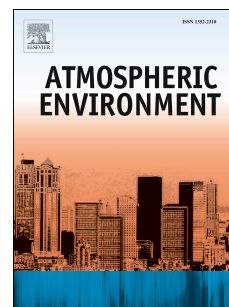


# Accepted Manuscript

Human health risk due to variations in PM<sub>10</sub>-PM<sub>2.5</sub> and associated PAHs levels

Beatriz S. Sosa, Andrés Porta, Jorge Esteban Colman Lerner, Roxana Banda Noriega, Laura Massolo



PII: S1352-2310(17)30232-7

DOI: [10.1016/j.atmosenv.2017.04.004](https://doi.org/10.1016/j.atmosenv.2017.04.004)

Reference: AEA 15270

To appear in: *Atmospheric Environment*

Received Date: 20 December 2016

Revised Date: 26 March 2017

Accepted Date: 3 April 2017

Please cite this article as: Sosa, B.S., Porta, André., Colman Lerner, J.E., Banda Noriega, R., Massolo, L., Human health risk due to variations in PM<sub>10</sub>-PM<sub>2.5</sub> and associated PAHs levels, *Atmospheric Environment* (2017), doi: 10.1016/j.atmosenv.2017.04.004.

This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

## Human health risk due to variations in PM<sub>10</sub>-PM<sub>2.5</sub> and associated PAHs levels

### Authors:

Beatriz S. Sosa<sup>ab</sup>, Andrés Porta<sup>cd</sup>, Jorge Esteban Colman Lerner<sup>be</sup>, Roxana Banda Noriega<sup>ad</sup>, Laura Massolo<sup>c</sup>

<sup>a</sup> Centro de Investigaciones y Estudios Ambientales, Universidad Nacional del Centro de la Provincia de Buenos Aires, Tandil, Argentina (CINEA- UNICEN). Paraje Arroyo Seco s/n CP: 7000. [beatrizsosa33@gmail.com](mailto:beatrizsosa33@gmail.com) TE +54 9 249 4385750

<sup>b</sup> Consejo Nacional de Investigaciones Científicas y Técnicas.

<sup>c</sup> Centro de Investigaciones Medioambientales, Facultad de Ciencias Exactas. Universidad Nacional de La Plata (CIMA-EXA-UNLP). 47 y 115. La Plata 1900. TE +4 221 4229329

<sup>d</sup> Comisión de Investigaciones Científicas de la Provincia de Buenos Aires.

<sup>e</sup> Centro de Investigaciones y desarrollo en ciencias Aplicadas “Dr. Jorge Ronco” (CINDECA-CONICET). 47 #257 La Plata 1900 54-221-421-1353

# Human health risk due to variations in PM<sub>10</sub>-PM<sub>2.5</sub> and associated PAHs levels

## Abstract.

WHO (2012) reports that chronic exposure to air pollutants, including particulate matter (PM), causes the death of 7 million people, constituting the most important environmental risk for health in the world. IARC classifies contaminated outdoor air as carcinogenic, Group 1 category. However, in our countries there are few studies regarding air pollution levels and possible associated effects on public health.

The current study determined PM and associated polycyclic aromatic hydrocarbons (PAHs) levels in outdoor air, identified their possible emission sources and analysed health risks in the city of Tandil (Argentina). PM<sub>10</sub> and PM<sub>2.5</sub> samples were collected using a low volume sampler (MiniVol TAS) in three areas: city centre, industrial and residential. Concentrations were determined by gravimetric methods and the content of the US EPA 16 priority PAHs was found by high performance liquid chromatography (HPLC). Description of the main emission sources and selection of monitoring sites resulted from spatial analysis and the IVE (International Vehicle Emissions) model was used in the characterisation of the traffic flow. Median values of 35.7  $\mu\text{gm}^{-3}$  and 9.6  $\mu\text{gm}^{-3}$  in PM<sub>10</sub> and PM<sub>2.5</sub> respectively and characteristic profiles were found for each area. Local values PAHs associated to PM<sub>10</sub> and PM<sub>2.5</sub>, in general, were lower than 10  $\text{ngm}^{-3}$ . The estimated Unit Risk for the three areas exceeds US EPA standards ( $9 \times 10^{-5}$ ). The number of deaths attributable to short term exposure to outdoor PM<sub>10</sub> was 4 cases in children under 5 years of age, and 21 cases in total population, for a relative risk of 1.037.

**Keywords:** Air quality; PM; PM-PAHs; Environmental burden of disease; Unit Risk; Tandil Argentina.

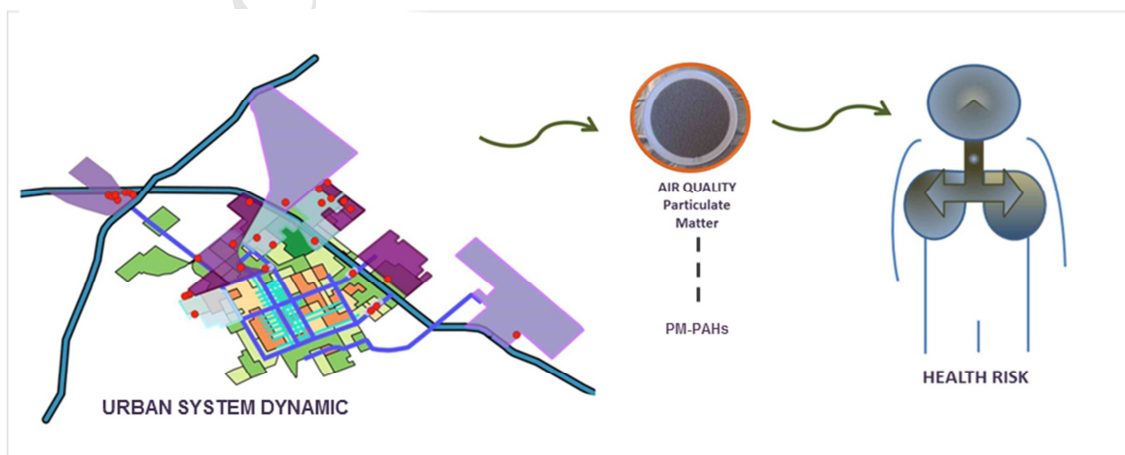
## Highlights

Urban PM and PAHs from point and mobile sources represent a health risk for citizens.

Pollutant characterisation according to land use areas facilitates the study of the sources involved and provides useful data for management.

Health risk due to PM exposure is a state indicator (diagnosis) of ambient air which allows to evaluate the relationship between anthropogenic activities and the deterioration of air quality

## Graphical Abstract



## 1. Introduction

Globally, 7 million deaths were attributable to the joint effects of household and ambient air pollution in 2012 (WHO, 2014). Recently, the International Agency for Research on Cancer (IARC) has classified the mixture of substances in outdoor air as carcinogenic to humans (Group 1). Although the composition of air pollution and exposure levels can vary considerably between different sites, the conclusions of the Working Group apply to all regions of the world. Particulate matter (PM) as the main component of outdoor air pollution, was evaluated separately and classified as carcinogenic to humans (IARC / WHO, 2013). Previous studies both at city and regional levels have shown that there is a strong link between exposure to coarse particulate matter  $PM_{10-2.5}$ , and mortality (Zanobetti, and Schwartz, 2009; Perez *et al.*, 2009; Schwartz *et al.*, 1996; Cifuentes, 2000; Pérez, 2010, US EPA, 2009a, WHO, 2013). Among the contaminants associated to PM, PAHs are considered parameters of interest because of their ubiquity, many sources of origin, and the presence of compounds which represent a health risk such as Benzo(a)pyrene (BaP). BaP is the reference toxic substance to study health risk from exposure to PAHs; it is used as an indicator of the relative toxicity of other PAHs in air mixtures (Nisbet and LaGoy, 1992, US EPA, 1986). As a result of epidemiological studies by WHO and US EPA, BaP is considered an indicator of carcinogenic risk associated with PAHs in air (Bostrom *et al.*, 2002; Collins *et al.*, 1998).

These compounds are found in the vapour phase or associated to suspended PM (particle phase) due to their low vapour pressure (especially with 5 or more rings). Its presence in PM correlates with particle size, finding the highest levels in the inhalable type. They are formed by incomplete combustion of organic matter. The main emission sources are industrial processes, road traffic, biomass combustion and domestic. Road traffic emissions are usually one of the most important sources in urban areas (Guevara, 2016; Pufulete, 2004, Fang *et al.*, 2010). Considering that WHO states that the presence of PM is a problem in both big and small cities, this study seeks to characterize suspended PM in outdoor air of a medium-sized city, Tandil, Argentina. The health risk from exposure to  $PM_{10}$  and PAHs associated to  $PM_{10}$  and  $PM_{2.5}$  are calculated as well.

The city is a growing urban location, with historical industrial activity in the town centre which has allowed both local development and an important vehicle fleet compared to its population size. With approximately 123,871 inhabitants in the district and a 50 km<sup>2</sup> area, Tandil is located in the mountain system of Tandilia which extends in a NW-SE direction, and constitutes a major tourist attraction. Foundry is one of the main industrial activities; however atmospheric emissions from these industries are subjected to poor environmental control measures (Sosa *et al.*, 2013a). This activity, together with the increasing vehicular fleet, poses a health risk to citizens.

This work is the first urban study which includes the diagnosis and characterisation of PM and associated PAH for the city of Tandil.

## 2. Materials and methods

### 2.1 Emission sources and monitoring sites

Several activities which are carried out in Tandil contribute to the deterioration of urban air quality. Following the methodology applied by other authors (Jamhari *et al.*, 2014; Massolo *et al.*, 2009), three monitoring sites were used: industrial (I), city centre (C), and residential or control (R). Both the identification of main emission sources and selection of monitoring sites were performed using spatial analysis tools in Quantum GIS Browser, with input from primary and secondary data on the activity of stationary and mobile sources (Figure 1). The analysis included the following variables:

*For point sources:* land uses established by the Land Management Plan for Tandil which classifies land use based on urban anthropogenic activities; spatial distribution of foundry industries and their characteristics (according to Sosa *et al.* 2013a and Sosa *et al.* 2013b); and, population density.

*For mobile sources:* Traffic flow in areas of interest. Traffic observation was made during autumn 2013 in three different areas based on the Regional Development Plan regulations on road transects. The counting method used followed the International Vehicle Emissions Model (IVEM) developed by International Sustainable Systems Research Centre (ISSRC) and the University of California Riverside (UCR) (IVE, 2008 - Attachment D). In order to know the general traffic distribution, vehicles were classified into seven categories: car, taxi / remise (private car service), vans, medium and large buses, light and heavy commercial vehicles, lorries, 2- and 3- wheel vehicles. Measurements were taken on working week days with normal urban activity and one day with part-time activity.

## **2.2 PM characterisation: sampling, extraction and analysis**

Atmospheric PM samples were taken from three areas indicated above: I C R. A low-volume portable sampler (Minivol TAS) with a volumetric flow of 5 L min<sup>-1</sup> was used during 120 h. Particulate matter was collected on polytetrafluoroethylene (PTFE) and fibreglass (46.2 mm in diameter). They were weighted before and after sampling under standardized conditions. Particulate matter concentrations of each sample were determined by working out the ratio between this gravimetric difference and the total volume of air passing through the filter. Samples of both types of particulate matter, inhalable (particle size <10 µm) and respirable (particle size <2.5 µm) were obtained using continuous but not simultaneous sampling methods (Baldauf *et al.*, 2001; Colman Lerner *et al.*, 2015; Colman Lerner *et al.*, 2016; Sosa, 2015)

PAHs content was analysed by high performance liquid chromatography (HPLC) with fluorescence detection. 16 US EPA priority PAHs were selected: naphthalene (Naph), acenaphthylene (Acpy), acenaphthene (Acp), fluorene (Flu), phenanthrene (Pha), anthracene (Ant), fluoranthene (Fl), pyrene (Pyr), benzo(a)anthracene (BaA), chrysene (Chr), benzo(b)fluoranthene (BbF), benzo(k)fluoranthene (BkF), benzo(a)pyrene (BaP), indeno(1,2,3-cd)pyrene (Ind), dibenzo(a,h)anthracene (DBA), benzo(g,h,i)perylene (BghiP). PAHs were extracted from each filter with hexane by sonication (TESTLAB TB10 ultrasonic bath, Power = 400W, frequency = 40kHz) for a period of 60 minutes. Then it was dried and 1ml of acetonitrile was added (J: Baker, HPLC quality). The recovery factor for each analyte was determined by adding a known amount of the 16 PAHs (Alltech, in triplicate) to unused filters and removing them by the same method, their values are between 0.714 and 1.015 except for anthracene (0.589). The obtained extracts, and blank samples obtained from the treatment of unused filters in triplicate, were analysed by high resolution liquid chromatography (UHPLC Nexera, Shimadzu) with PDA detectors (diode array technology) and fluorescence. Chromatographic conditions used: C18 column (Zorbax Eclipse PAH 100mm x 4.6mm, 3.5µm), acetonitrile isocratic mode (40%) – water (60%) during 0.66 minutes and then linear gradient up to 100% acetonitrile in 20 minutes at a 2.0 L min<sup>-1</sup> flow. PAHs were quantified with UV (PDA detector, 220nm) and fluorescence: 280 nm excitation and 425 nm, emission. The values corresponding to the detection limits for each PAH are included in the range 0.012 to 1865 mgL<sup>-1</sup>, except naphthalene with 2.903 mgL<sup>-1</sup>. Individual values of recovery factors and detection limits for any PAHs are shown in relation to tables B1 and B2 in the supplementary materials (NIOSH 5506, 1998; Colman Lerner *et al.*, 2015; Colman Lerner *et al.*, 2016).

PM sampling was made in autumn / winter (between March 2013 and September 2014) to ensure the most unfavourable conditions, since high relative humidity and low wind speeds prevail in these seasons (Arkouliet *et al.*, 2010; Kukkonen *et al.*, 2005; Massolo *et al.*, 2002; Picone *et al.*, 2014; Vardoulakis and Kassomenos, 2008). In order to analyse the influence of

meteorological factors, temperature (°C), relative humidity, rainfall and wind speed were observed and recorded for a period of 60 days which included, the sampling period, previous and subsequent days. Data was supplied by the Aero Station, National Weather Service (Located 18 km from the city, northeast direction).

### 2.3 Health risk due to PM exposure

#### 2.3.1 Environmental Burden of Disease (EBD) for mortality due to short-term exposure in children

The method adopted for calculating the expected number of mortality cases due to air pollution was proposed by the World Health Organization (Ostro, 2004).

Based on available evidence, a reasonable estimate of the Environmental Burden of Disease (EBD) for mortality due to short-term exposure is generally a 0.6% to 1% increase per  $10\mu\text{gm}^{-3}$  PM; and in children the increase is generally a 1.66% (95% CI= 0.34-3.0) per  $10\mu\text{gm}^{-3}$ . This assumption is applicable to all-cause mortality. This range reflects the evidence from a variety of cities and average times (including single and multi-day lags). To quantify this effect, the relative risk (RR) can be specified using Equation 1 (Ostro, 2004).

$$\text{RR} = \exp[\beta (X - X_0)] \quad [\text{Equation 1}]$$

Where:

$\beta$  = 0.00166, suggested  $\beta$ coefficient (95% CI) to subgroup age <5 years; or, 0.0008 proposed best estimate to all age (range 0.0006 – 0.0010).

X = current annual mean  $\text{PM}_{10}$  concentration ( $\mu\text{gm}^{-3}$ )

$X_0$  =  $10\mu\text{gm}^{-3}$ , WHO suggested background concentration to short-term exposure.

#### Calculating the disease burden

Once the relative risk was determined, the Impact Fraction (AF) of health effects from air pollution for the exposed population (children <5 years) was calculated using Equation 2.

$$\text{AF} = (\text{RR} - 1) / \text{RR} \quad [\text{Equation 2}]$$

Where:

AF= proportion of the environmental burden of disease attributable to  $\text{PM}_{10}$

#### Calculating the expected number of mortality cases due to air pollution

Finally, to calculate the expected number of mortality cases due to air pollution (E), the AF was applied to the total number of deaths in Equation 3.

$$\text{E} = \text{AF} \times \text{B} \times \text{P} \quad [\text{Equation 3}]$$

Where:

E = expected number of deaths due to outdoor air pollution.

B = incidence of the given health effect on the population (deaths per 1000 people).

P = relevant exposed population for the health effect.

#### 2.3.2 Calculating Toxic Equivalency Factors – TEF.

BaP toxic equivalents (BaP<sub>eq</sub>) were found by the method defined by Nisbet and LaGoy, in 1992; which is still valid and implemented by the National Pollutant Inventory- only replacing the PAHs concentration by its mass (NEPC, 2006).

Chemical compounds that exert their toxicity by the same mechanism may be expressed in relation to a member of that group (in this case BaP), which in general is the most toxic. This is known as Toxic Equivalency Factor (TEF).

BaP<sub>eq</sub> is obtained from multiplying the concentration of the compound by the corresponding TEF, while the sum of BaP<sub>eq</sub> ( $\Sigma$ BaP<sub>eq</sub>) allows knowing the total equivalent toxic concentration of the mixture (Vargas *et al.*, 2013; Nisbet and LaGoy, 1992; NEPC, 2006).

TEFs values for each compound are shown in Table 9 together with the results.

### **2.3.3 Estimated risk associated with PAHs based on Unit Risk.**

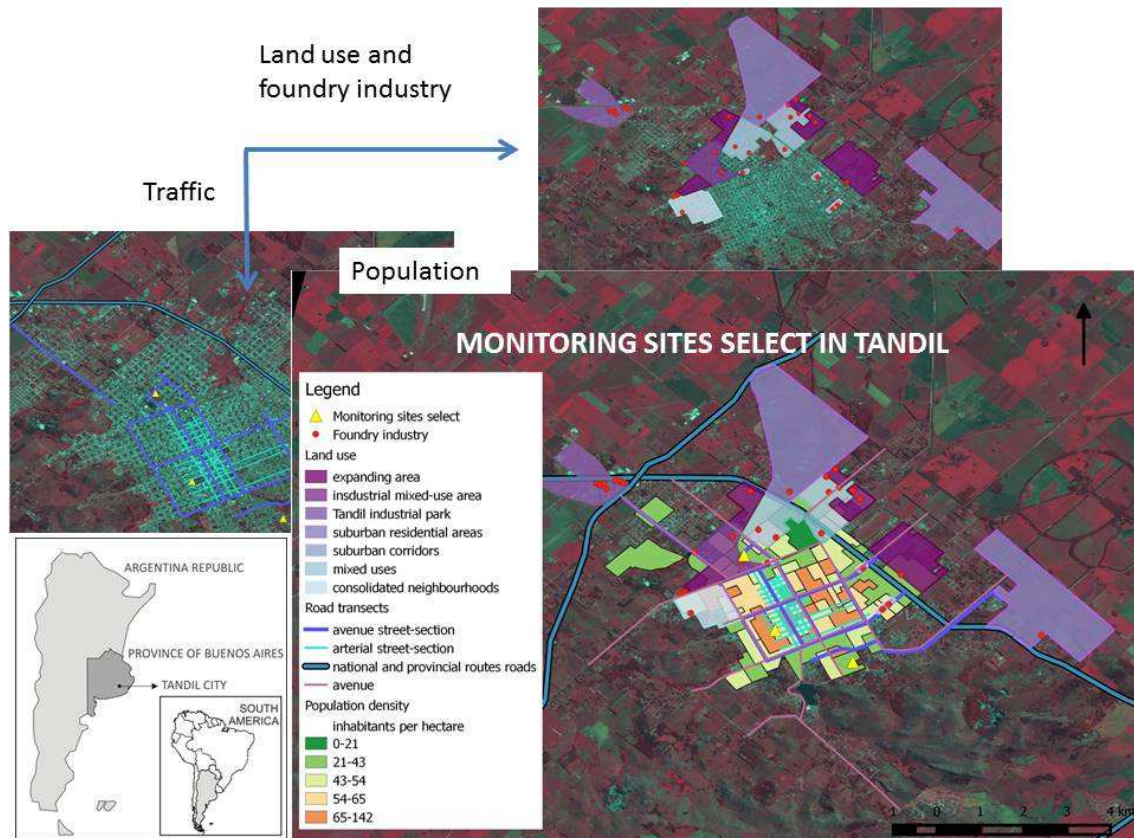
The Unit Risk model (WHO, 2000), which refers to the cancer rate a contaminant can cause for a given exposure level, was used in this study. To estimate the increased risk derived from PAHs exposure, the linearized multistage model of WHO was used. This model provides risk estimates in cases of extrapolation of risks with low dose and calculates levels associated with excess cancer risk of 1: 10 000, 1: 100 000 and 1: 1 000 000. Estimates are made from the additional risk of developing cancer throughout life in a hypothetical population whose individuals are continuously exposed from birth to a concentration of 1  $\mu\text{gm}^{-3}$  agent in the air they breathe (WHO; 2000). On this basis, the increased risk is expressed for  $1 \times 10^{-6}$ ,  $1 \times 10^{-5}$ ,  $1 \times 10^{-4}$  (1  $\text{ngm}^{-3}$  corresponds to a risk of  $1 \times 10^{-4}$ ).

## **3. Results and discussion**

### **3.1 Emission sources and monitoring sites**

The selection of monitoring sites resulted from the analysis of the two main emission sources in the city, the casting industry and motor vehicles. Previous studies characterised casting activities (Sosa *et al.*, 2013a,b) as the contribution of each industry from the monthly production and technologies used to control emissions. This work makes a spatial analysis of this data, considering the predominant land uses and traffic characteristics to identify the best sampling sites for I C R (Figure 1).





**Figure 1. Selection of monitoring areas: industrial (I), urban (city, C) and residential (R).**

The vehicle counting was carried out in the local urban centre with predominant commercial and administrative land use (area C), next to the larger metallurgical industry within the industrial area (area I), in places with very low number of vehicles and residential land use (area R). The assessment of traffic flow related to urban activity at different times of the day, showed that the average number of vehicles passing through I C R (Table 1) was 148 / 20 min at peak hour (18 to 20 pm) and 112 / 20 min at off-peak hour (14 to 16 pm). In comparative terms, the minimum number of vehicles passing through a site at peak hour was twice that for non-peak hour; however, when considering maximum possible values the difference is not so significant.

**Table 1. Traffic Flow: descriptive statistical results.**

| Traffic Flow  | n  | Mean   | $\sigma$ | %CV   | Min    | Max | Median | Q1     | Q3     |
|---------------|----|--------|----------|-------|--------|-----|--------|--------|--------|
| Peak hour     | 18 | 148.06 | 24.33    | 16.43 | 104.00 | 195 | 152.00 | 125.00 | 164.00 |
| Off-peak hour | 18 | 112.39 | 32.25    | 28.70 | 52.00  | 178 | 114.00 | 96.00  | 133.00 |

$\sigma$ : standard deviation; %CV: coefficient of variation expressed as a percentage.

The proportion of vehicles by category in each area (Table 2) resulted in no major differences in terms of fleet distribution; however, the total number of vehicles increases and tends to double between R and I, and between I and C.



**Table 2. Vehicle distribution by categories in three areas of Tandil, I C R 2013.**

| VehicleCategory                   | I    |      | C    |      | R    |      |
|-----------------------------------|------|------|------|------|------|------|
|                                   | N    | %    | n    | %    | n    | %    |
| Car                               | 1353 | 63.8 | 3042 | 65.8 | 609  | 55.7 |
| Taxi/ Remis (private car service) | 102  | 4.8  | 293  | 6.3  | 35   | 3.2  |
| Vans                              | 368  | 17.3 | 676  | 14.6 | 252  | 23.0 |
| Buses                             | 46   | 2.2  | 124  | 2.7  | 17   | 1.6  |
| Commercialvehicles                | 81   | 3.8  | 207  | 4.5  | 81   | 7.4  |
| Lorries                           | 45   | 2.1  | 36   | 0.8  | 35   | 3.2  |
| Two- and Three- Wheel vehicles    | 127  | 6.0  | 246  | 5.3  | 65   | 5.9  |
| Total                             | 2122 | 100  | 4624 | 100  | 1094 | 100  |

n= Total number of vehicles in 36 twenty-minute counts (peak + off-peak hours).

Analysis of these findings resulted in the selection of the three monitoring sites I C R (Figure 1).

### 3.2 Characterisation of PM air quality

#### 3.2.1 PM<sub>10</sub> and PM<sub>2.5</sub> concentrations

Table 3 shows PM concentrations measured in the three urban areas in 2013 and 2014. Due to the presence of high concentrations in winter 2013 and autumn 2014 campaigns -for I and C respectively- and the significant impact of the smaller particles on health, it was decided to carry out the autumn / winter 2016 campaign for PM<sub>2.5</sub> with more samples at each site.

**Table 3. PM<sub>10</sub> and PM<sub>2.5</sub> concentrations in three urban areas in Tandil.**

| Area                       | Autumn |      | Winter | Winter/Spring | Autumn/Winter        |       | Median (µgm <sup>-3</sup> ) |
|----------------------------|--------|------|--------|---------------|----------------------|-------|-----------------------------|
|                            | 2013   | 2014 | 2013   | 2014          | 2016                 |       |                             |
| PM10 (µgm <sup>-3</sup> )  |        |      |        |               |                      |       |                             |
| I                          | 32.6   | 57.6 | 49.1   | --            | --                   | --    | 49.1                        |
| C                          | 17.4   | 24.8 | <0.1   | 29.4          | --                   | --    | 24.8                        |
| R                          | 37.7   | 41.7 | 35.7   | --            | --                   | --    | 37.7                        |
| PM2,5 (µgm <sup>-3</sup> ) |        |      |        |               |                      |       |                             |
| I                          | 3.4    | 11.5 | 31.7   | 14            | 8.5<br>9<br>9.6      | 9*    | 11.5                        |
| C                          | <0.1   | 27   | 9.6    | 8.1           | 14.9<br>18.9<br>31.8 | 21.9* | 15.75                       |
| R                          | 7      | 9.6  | <0.1   | 6.2           |                      | --    | 7                           |

\*PM  $\bar{X}$  for autumn/winter 2016.

The concentrations found are, in turn, the result of the climatic conditions during those periods. Although samples were not taken simultaneously, not be found net differences between the values of relative humidity, temperature and wind speed, the factors that affect atmospheric PM concentrations.

However, it was observed that wind speeds above 20 km / h had a significant impact, resulting in atypical values of PM, for example in winter PM<sub>10</sub> / C and in PM<sub>2.5</sub> / R; or autumn 2013 PM<sub>10</sub> / C. Wind speed was also responsible for concentrations of lower levels of PM<sub>2.5</sub> in area I.

Considering the urban system dynamics in the study of PM concentrations, information about human activities within the city is required to make comparisons between I C R. High values in the industrial area correlate with the sources found there, as this is the area where the highest local metalworking production takes place (Sosa *et al.*, 2013a), and shows a similar peak hour traffic flow to the central area. Several authors associate high levels of PM<sub>10</sub> and PM<sub>2.5</sub> to combustion sources and traffic congestion in central urban areas in particular (Figuerelo y Dávila, 2005:68; Lewtas, 2007; Markakis, *et. al.*, 2010; WHO, 2013).

Spatial observation of PM concentrations shows the great incidence of the mobile source, evidencing the general association of PM<sub>2.5</sub> with anthropogenic combustion sources. The highest PM<sub>2.5</sub> mean values were measured in the city centre, explained by the large number of vehicles during high urban activity, with a significant percentage of diesel vehicles, that make up a 54.3% the whole vehicle fleet in Tandil, while in Argentina these proportion is 35% (Sosa, 2015). Approximately, 92% of diesel engine particle emissions are less than 1µm in diameter. Exposure to this mixture of chemicals should be reduced worldwide because they are harmful to human health (NTP, 2016; WHO 2012).

Higher values in R compared to I and C can be attributed to a significant burden of dust from the bare soil in the area while PM<sub>2.5</sub> median concentrations in I and C evidence the significant contribution of anthropogenic sources.

In a regional analysis (Table 4), values for areas I, C and R have been recently studied in cities with different population size, industrial activity and traffic flow: in Tandil, I is occupied by small and medium industries whereas in Bahía Blanca and La Plata it houses large petrochemical centres. The Autonomous City of Buenos Aires (CABA) is the most complex, for its size and in terms of human activity and emission sources.

**Table 4. PM<sub>10</sub> and PM<sub>2.5</sub> concentrations in regional urban areas.**

| Area                                    | Tandi <sup>1</sup><br>123.874 inhab. | Bahía Blanca <sup>2</sup><br>301.572 inhab. | CABA <sup>3</sup><br>2.890.151 inhab. | La Plat <sup>2</sup><br>654.324 inhab. |
|-----------------------------------------|--------------------------------------|---------------------------------------------|---------------------------------------|----------------------------------------|
| PM <sub>10</sub> (µg m <sup>-3</sup> )  |                                      |                                             |                                       |                                        |
| I                                       | 49.1                                 | 117.1                                       | --                                    | 62.0                                   |
| C                                       | 24.8                                 | 138.1                                       | 35.0                                  | 31.5                                   |
| R                                       | 37.7                                 | 52.9                                        | 45.0                                  | 33.8                                   |
| PM <sub>2.5</sub> (µg m <sup>-3</sup> ) |                                      |                                             |                                       |                                        |
| I                                       | 11.5                                 | 105.5                                       | --                                    | 33.7                                   |
| C                                       | 15.75                                | 58.0                                        | 16.0                                  | 18.0                                   |
| R                                       | 7                                    | 32.0                                        | 20.0                                  | 18.2                                   |

Source: <sup>1</sup>Elaborated by the authors. <sup>2</sup>Colman Lerner, 2012. <sup>3</sup>Arkouli, 2010.

Note: Given the heterogeneity of air and reduced data values in Tandil, Bahía Blanca and La Plata, medium values are used. Mean values are used for CABA.

PM values for Bahía Blanca are quite different from other cities in the region. This is likely to happen due to the significant contribution made by dust blown from the barren soil by the wind. This natural situation is aggravated in PM<sub>2.5</sub> by the important contribution of its industrial park (fixed sources + mobile sources).

Regarding PM<sub>10</sub>, except for Bahía Blanca, regional and city values are in the same order.

When comparing inner city areas, some remarkable similarities can be observed. In some cases population size explains the similar values found in very different areas between cities (R in CABA, I in Tandil). In other cases, further studies are required to provide more knowledge about each urban dynamics. This will help to explain similar values for C in CABA, La Plata and Tandil, although there is no doubt that the mobile source is the main contribution.

At local scale, the median concentrations of PM<sub>10</sub> and PM<sub>2.5</sub> found in Tandil during the study period were 35.7 µgm<sup>-3</sup> and 9.6 µgm<sup>-3</sup> respectively. In order to protect health and the environment, WHO, EU and US EPA have set annual PM standards: 20 µgm<sup>-3</sup> and 40 µgm<sup>-3</sup> for PM<sub>10</sub> (WHO and EU, respectively), and 10 µgm<sup>-3</sup>, 25 µgm<sup>-3</sup>, 12 µgm<sup>-3</sup> for PM<sub>2.5</sub> (WHO, 2006; Directive 2008/50 / EC; Federal Register US EPA, 2013, respectively). Comparing this standard, with the values in Tandil, it is evident that median PM<sub>2.5</sub> ranges below the maximum guideline value set by WHO, EU and US EPA. However, there are individual values closer to EU standard. Regarding PM<sub>10</sub> in Tandil, its median values are between the WHO and EU standards.

Moreover, in comparison with other values from different places in the world, PM<sub>2.5</sub> in Tandil (9.6 µgm<sup>-3</sup>) is positioned well below most cities and countries. In addition PM concentrations are similar to those found in cities in Brazil, Spain and Germany. However, they are very close to the maximum values allowed in Canada. Regarding PM<sub>10</sub> concentrations, Latin American and Asian cities still need to improve air quality in order to reach annual mean values proposed by the agencies mentioned above. Although Tandil is an intermediate city, its PM<sub>10</sub> concentration is similar to those in other cities with populations of one to two higher orders of magnitude (e.g. Buenos Aires, London, Frankfurt). Details of the comparison between values found in this study and in other cities around the world are provided in the Supplement Material, Table A1 and A2.

Retaking the local analysis, it was found that PM<sub>10</sub> concentrations in area I exceed EU recommended values, however PM<sub>2.5</sub> is half below the threshold. Compared to WHO standards, PM<sub>10</sub> median value is almost twice, but PM<sub>2.5</sub> is closer to the recommended value. In area C, both PM values fit the EU thresholds. According to WHO standard both PM<sub>10</sub> and PM<sub>2.5</sub> exceed its recommended values (PM<sub>2.5</sub> 50% higher). Clearly, the contribution of the vehicular source is determinant in this area. Finally, R is within the EU thresholds for both PM sizes analysed; however, PM<sub>10</sub> is almost twice as recommended by WHO, and PM<sub>2.5</sub> is far below the limit set by this organisation.

It is important to consider the local standards in the Province of Buenos Aires, which sets maximum annual means for PM<sub>10</sub> only. Except for area I in autumn 2014, overall median values, means for each of the three urban areas, and individual samples do not exceed the reference value of 50µgm<sup>-3</sup>.

### **3.2.2 PM- associated PAHs.**

Some of the sixteen priority PAHs selected are considered to be possible or probable human carcinogens, and have been listed also by the International Agency for Research on Cancer (IARC) as priority pollutants, these include the Groups 1 carcinogenic to humans (BaP), 2A probably carcinogenic to humans (DBA), 2B possibly carcinogenic to humans (BaA; Chr; BbF, BkF;Ind), and 3 not classifiable as to its carcinogenicity to humans (Acp,Flu, Pha,Ant, Fl,Pyr, BghiP) (IARC, 2014); the acenaphthylene (Acpy) is not classified.

For the autumn 2013 – autumn 2014 period, values of Naph, Acpy, Acp, DBA and Flu were found below detection limits in all cases; same for Pha, Pyr, Chr in PM<sub>10</sub> and Ant in PM<sub>2.5</sub>.

An increase of PAHs in PM<sub>10</sub> total values could be noticed in the industrial area for the last period as a result of high levels of Ant in autumn 2014 (see supplementary material, Table B1) This compound is used for the manufacture of fast dyes, as a diluent for wood preservatives and in the production of synthetic fibres, plastics and monocrystals; it is associated mainly to the combustion of gasoline, diesel or wood burning, all these corresponding to inputs of manufacturing processes developed in the area. These origins explain the presence of PAHs in all other samples, in relation to vehicular activity and burning of wood for heating, mainly in areas C and R. Total PAHs in R are twice the values found in C. (Carrasquero-Durán, 2012;

Mastandrea, *et al.*, 2005; Wiriyaet *et al.*, 2013) 28.8% of the PAHs present in these samples have been classified by IARC as compounds of interest due to their health effects on humans. BaP, the main carcinogenic compound indicator, has been found at its highest concentration in area I and at its lowest in R. These values are consistent with other studies carried out in nearby areas (Massolo, 2004).

With reference to mean concentrations found in PM<sub>2.5</sub> as shown in Table B2 (supplementary material), 35.5% are carcinogenic. When considering health risk, this is even more significant because particulate matter of such a small size enters the most sensitive areas of the respiratory system. This value is affected by high concentrations of Ind in one of the samples (winter 2013), however this high value cannot be attributed only to the presence of this compound as the mean of the remaining compounds is equally significant (31.19%) when Ind is not considered. The concentration of Pyr in autumn 2013 reveals the influence of human activity in C. This measurement was made from March 24<sup>th</sup> to 31<sup>st</sup>, Easter week holiday, a significant celebration in Tandil when its population almost doubles mainly between Holy Thursday and Easter Sunday. Consequently C showed greater presence and concentration of PAHs compared to I and R during that campaign. Studies carried out in Asia and reviewed by Chang and others have associated Pyr with vehicular traffic (Chang *et al.*, 2006; Kong, *et al.*, 2015).

In an overall reflection on PAHs (PM<sub>2.5</sub> and PM<sub>10</sub>) and its related sources, the results of this study coincide with the findings of other authors who associate AcP and, Fl, Flu, Pha, Pyr, Chr, emissions from vehicle exhausts. However the predominance of BaP and BaN highest values occurred in area C, although authors associate these compounds with industrial sources (Chang *et al.*, 2006; Massolo, 2004; Marino, 2011; Astoviza, 2014; ATSDR, 1995; Fang *et al.*, 2006). Yang *et al.* (1998) relates emission sources to the number of rings of the contaminants and the foundry industries in particular, agreeing that BaP and BaA come from industrial chimneys. Notably, BaP estimated values in urban air in Europe are in the range of 1 to 10 ngm<sup>-3</sup> (WHO 2000), while concentrations found in PM<sub>10</sub> and PM<sub>2.5</sub> in Tandil, in general, are below these figures.

Meteorological factors during the sampling period support the conclusion that these (winter low temperatures and low light intensities) are not enough to influence the chemical stability of PAHs associated to particulate matter due to photolysis (Dimshkiet *et al.*, 2001, Chang *et al.*, 2006). During his work in the south of the River Plate Basin, Astoviza (2014) found that the absence of a PAHs consistent trend over time could be related to the multiplicity and continuity of the emission sources (fixed and mobile) and to the fact that the area under study has no significant weather changes; although few samples were taken in Tandil, the results of this research are consistent with this. In this sense the importance of sampling in the colder seasons is justified from the predominance of carcinogenic PAHs in finer particles. Analyses in La Plata (Argentina) and Leipzig (Germany), found that the concentration of carcinogenic PAHs in the finest fraction is 10 times higher in winter than in summer, except for industrial areas where values are equally high (Massolo *et al.*, 2002).

Carcinogenic PAHs analysis in relation to total PAHs concentrations in PM<sub>10</sub> and PM<sub>2.5</sub> samples are consistent with those from other authors (Chang *et al.*, 2006; Fang *et al.*, 2006.), with higher proportions of carcinogenic compounds in fine particles (PM<sub>2.5</sub>). However the difference between proportions for different particle sizes is smaller than those reported by these authors.

### 3.3 Health risk due to PM exposure

#### 3.3.1 Environmental Burden of Disease (EBD) for mortality due to short-term exposure in children

WHO (2006) suggests that the health risks associated with short-term exposures to  $PM_{10}$  are likely to be similar in cities in developed and developing countries, producing an increase in mortality of around 0.5% for each  $10 \mu g m^{-3}$  increase in the daily concentration.

**Table 5. Expected number of mortality cases due to air pollution ( $PM_{10}$ ) by short-term exposure in Tandil.**

| X                | $X_0$ | B      | Relative risk | Impact Fraction | Population          | Population incidence | E     | Cases |
|------------------|-------|--------|---------------|-----------------|---------------------|----------------------|-------|-------|
| Children 0-4 age |       |        |               |                 |                     |                      |       |       |
| 32.6             | 10    | 0.0016 | 1.037         | 0.036           | 8855 <sup>1</sup>   | 11.6 <sup>3</sup>    | 3.65  | 4     |
| All ages         |       |        |               |                 |                     |                      |       |       |
| 32.6             | 10    | 0.0008 | 1.018         | 0.018           | 123871 <sup>2</sup> | 9.5 <sup>4</sup>     | 21.08 | 21    |

<sup>1</sup>Exposed infant population ((INDEC, 2010).

<sup>2</sup>Tandil population, all ages (INDEC, 2010).

<sup>3</sup>Infant mortality rate Tandil (MSAL, 2012).

<sup>4</sup>Overall mortality rate, Tandil (MSAL, 2012).

These results provide a first approach to cases of deaths in the observed populations, which can be used to set targets for reducing the number of cases; however, it must be noted that this methodology makes estimates of mortality for the short term from acute respiratory diseases but it does not identify particular causes of death. To improve this analysis, access to information in sufficient quantity and quality is critical for the discrimination and adjudication of mortality cases by type of respiratory disease over the total of these diseases. This would greatly facilitate a more precise study of health effects from exposure to  $PM_{10}$ .

### 3.3.2 Toxic Equivalency Factors – TEF.

Tables C1 and C2 (supplementary material) show results for  $BaP_{eq}$  y  $\sum BaP_{eq}$  during this study considering maximum  $PM_{10}$  and  $PM_{2.5}$  concentrations in the sampling areas (I, C, R).

In  $PM_{10}$ , BaP carcinogenic contribution for the mixture in all areas ranges between 22.9% and 25.9%, with the highest values in areas I and R. Ind is the most representative compound in the PAHs mixture in area C, showing high values in the other two areas too.

BbF, BaP, DBA, Ind and Ant are the main compounds in the PAHs mixture, contributing with values between 87.42% and 89.29%, they seem to have a similar carcinogenic role with some variation by area, being the BbF the one with the weakest contribution. Ant carcinogenic potential is 10 times lower than BbF, however its contribution is higher in areas I and R. At this point, it is important to emphasize that the values taken for this analysis are the maximum concentrations recorded<sup>1</sup>, and in some cases coincide with the only observation above detection limit of the analytical sampling method. With more measurements, these results would strengthen BaP as the main indicator in the mixture. In any case, these five predominant PAHs correspond to those whose TEF has a high carcinogenic potential. In the analysis by areas, the sum of the BaPeq is similar for C and R while in I the values are twice as high.

In  $PM_{2.5}$  (Table C2) the four predominant PAHs in the mixtures: BbF, BaP, DBA and Ind, coincide with those found in  $PM_{10}$ ; with percentages ranging between 83 and 98.6%. BaP and Ind are the main contributors. The mixture of PAHs in R is characterised almost exclusively by

1 As in previous analyses, the *precautionary principle* criterion is adopted to choose the values.



the presence of Ind which contributes 93% of total. At this point the high values found for these PAHs, and their impact on BaP<sub>eq</sub> values are explained on the basis of the above for PM<sub>10</sub>, added to the particular situation of vehicular activity for the Easter celebration. To round up, the comparative analysis of  $\Sigma$ BaP<sub>eq</sub> for areas I, C and R results in a similar total toxic equivalent concentration in I and R, and much higher values in C.

### 3.3.3 PAHs associated risk based on Unit Risk

Retaking BaP<sub>eq</sub> concentrations in the areas under study, the increased individual lifetime cancer risk due to BaP and PAHs exposure for each area under study is shown in Table 10. Current literature provides several unit risk estimates for PAHs inhalation expressed as BaP. US EPA Toxicological Review (2013) states values of  $8.7 \times 10^{-5}$  for WHO (WHO, 2000) and California Environmental Protection Agency estimates a value of  $1.1 \times 10^{-5}$  (OEHHA, 1994); US EPA data base suggests a BaP Unit Risk of  $8.8 \times 10^{-7}$  (US EPA, 2009b); Europe Air Quality Guidelines set a  $9 \times 10^{-5}$  Unit Risk (Bostrom *et al.*, 2002). However the most widely used Unit Risk is the US EPA  $1 \times 10^{-6}$  for PAHs mixtures for the year 2000 (IPCS, 2000). Table D (see supplementary material) shows the PM<sub>10</sub> and PM<sub>2.5</sub> increased risk in the three areas: I, C, R.

These findings show that the majority of total values for PM<sub>10</sub> and PM<sub>2.5</sub> mixtures are near the  $9 \times 10^{-5}$  threshold considered acceptable by Bostrom but they exceed the US EPA  $1 \times 10^{-6}$  standard and, in all cases, the  $8.8 \times 10^{-7}$  threshold for PAHs mixtures. Area R has a higher increased risk in PM<sub>2.5</sub> by order of magnitude compared to the rest of the areas ( $\Sigma$ BaP  $3.51 \times 10^{-4}$ ) –we must consider that this area registered a high Ind measurement as explained earlier.

Retaking the point made in section 3.2, another possibility regarding the impact of air pollution on children's health is to analyze the levels of PM compared to levels guide for air quality of WHO or European legislation. Table 6 shows PM levels found in the three areas of Tandil in relation to levels above the guide.

**Table 6. Concentration of PM<sub>10</sub> compared to levels guide air quality by WHO and EU.**

| PM <sub>10</sub> range (µgm <sup>-3</sup> ) | Quality <sup>a</sup> | Pollution level <sup>a</sup> | Mortality Increment <sup>b</sup> | Experimental data Tandil                                                                                                 |
|---------------------------------------------|----------------------|------------------------------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 0-25                                        | Excellent            | Low                          |                                  | City center (17.92 µgm <sup>-3</sup> )                                                                                   |
| 25-50                                       | Good                 | Normal                       | 0                                | Industrial (46.43 µgm <sup>-3</sup> )<br>Residential (38.35 µgm <sup>-3</sup> )<br>Tandil city (32.6 µgm <sup>-3</sup> ) |
| 50-75                                       | Poor                 | High                         | 1.2 %                            |                                                                                                                          |
| 75-100                                      | Bad                  | Very High                    | 2.5 %                            |                                                                                                                          |
| 100-150                                     | Very Bad             | Very High                    | 5 %                              |                                                                                                                          |

a. European legislation (Vicente *et al.*, 2012)

b. WHO guidelines for Europe (WHO, 2000)

It can be seen that for PM<sub>10</sub>, levels are good for Industrial and Residential area; and excellent for City center. Tandil concentration is situated in the range of Good quality, with a 0% increase in infant mortality.

## 4. Conclusions

Strategic monitoring sites were determined according to the main urban emission sources. Vehicle fleet distribution by category keeps similar proportions in all the zones; however traffic flow is obviously much higher in C than in I and R. Similar PM<sub>2.5</sub> values found in C and I evidence the significance of the mobile source in the deterioration of urban air quality in areas with a marked commercial activity.

Taking into account WHO guidelines as the desirable levels (for  $PM_{10}$  and  $PM_{2.5}$ ), Tandil is well above the maximum WHO thresholds for mean annual concentrations but it is within values established by US EPA. Median concentrations of  $35.7 \mu g m^{-3}$  and  $9.6 \mu g m^{-3}$  for  $PM_{10}$  and  $PM_{2.5}$  respectively were found, with wind being the single meteorological factor showing a clear relationship with variations in these levels. The highest median  $PM_{10}$  levels were measured in the industrial area, followed by the residential area where the existing bare soil made a significant contribution. The highest  $PM_{2.5}$  levels were found in the industrial area, followed by the city centre due to the characteristic large vehicular fleet present in the area, particular situations of high urban activity and a high percentage of diesel vehicles.

PAHs such as naphthalene, AcPy, Acp and Flu were not detected in the samples. Pyr and Ind values were high in  $PM_{2.5}$ ; while Ant was high in  $PM_{10}$ . FL, PA, Pyr, Chr were associated to vehicle exhausts; BaP and BaA related to industrial mobile sources. BaP and BaA were predominant in area C, although other authors associate these indicators mainly to industrial sources. In general, local values PAHs associated to  $PM_{10}$  and  $PM_{2.5}$  were lower than  $10 ng m^{-3}$  (Supplementary material, Tables B1 and B2).

The number of deaths attributable to short term exposure to outdoor  $PM_{10}$  was 4 cases (AF 0.036) and 21 cases (AF 0.018) for a relative risk of 1.037 in children under 5 years of age and of 1.018 in total population respectively. The life-time risk of developing cancer by PAH exposure in the three areas under study was close to the  $9 \times 10^{-5}$  limit, and exceeding in all cases US EPA more strict values.

The incidence of the mobile source in the city centre and the percentage of carcinogen PAHs in fine particles would show that  $PM_{2.5}$  is the most relevant variable in this study and strengthens the importance of monitoring in cold stations. This study provides unique data, at a local scale, which constitutes both a starting point and baseline in the study of urban air quality. It also contributes to the diagnosis and characterization of air quality in the region.

## Acknowledgements

The authors wish to acknowledge the following institutions which contributed with funds for this study: the National Council of Scientific and Technological Research (CONICET), the Scientific Research Committee of the Province of Buenos Aires (CIC PBA), Universidad Nacional de La Plata (UNLP), Universidad Nacional del Centro de la Provincia de Buenos Aires (UNICEN).

## Bibliography

Arkouli, M.; Ulke, A. G.; Endlicher W.; Baumbach, G.; Schultz, E.; Vogt, U.; Muller, M.; Dawidowski, L.; Faggi, A.; Wolf-Benning, U.; Scheffknecht, G. 2010. Distribution and temporal behavior of particulate matter over the urban area of Buenos Aires. Atmospheric Pollution Research 1 (2010, Jan), pp. 1-8. <http://dx.doi.org/10.5094/APR.2010.001>

Astoviza, M. L. 2014. Evaluación de la distribución de contaminantes orgánicos persistentes (COPs) en aire en la zona de la cuenca del Plata mediante muestreadores pasivos artificiales. Tesis doctoral. Facultad de Ciencias Naturales y Museo. Universidad Nacional de La Plata.

ATSDR. 1995. Agency for Toxic Substances and Disease Registry. Toxicological profile for polycyclic aromatic hydrocarbons. Atlanta, GA: U.S. Department of Health and Human Services, Public Health Service.

- 476 Baldauf, R.W., Lane, D.D., Marotz, G.A., Wiener, R.W. 2001. Performance evaluation of the  
477 portable MiniVOL particulate matter sampler. *Atmospheric Environment* 35 (2001, December)  
478 pp- 6087–6091. **DOI:**10.1016/S1352-2310(01)00403-4.
- 479 Bostrom C.; Gerde, P.; Hanberg, A.; Jernstrom, B.; Johansson, C.; Kyrklund, T.; Rannug, A.;  
480 Tornqvist, M.; Victorin, K.; Westerholm, R. 2002. Cancer Risk Assessment, Indicators, and  
481 Guidelines for Polycyclic Aromatic Hydrocarbons in the Ambient  
482 Air. *Environmental Health Perspectives*, 110 (Supp 3) (2002), pp.451-88.
- 483 Carrasquero-Durán, A.; González Suárez, A. 2012. Determinación de naftaleno, fenantreno,  
484 antraceno y pireno en polvo de la calle de la ciudad de Maracay, Venezuela. *CIENCIA* 20 (2012),  
485 pp. 33-42.
- 486 Chang, K-F; Fang, G-C; Chen, J-C; Wu, Yuh-Shen. 2006. Atmospheric polycyclic aromatic  
487 hydrocarbons (PAHs) in ASIA: A review from 1999 to 2004. *Environment Pollution*, 142 (2006,  
488 Aug), pp. 388-396. **DOI:**10.1016/j.envpol.2005.09.025
- 489 Cifuentes L.A., Vega J., Köpfer K., Lave L.B. 2000. Effect of the fine fraction of particulate  
490 matter versus the coarse mass and other pollutants on daily mortality in Santiago, Chile. *J Air*  
491 *Waste Manag Assoc*, 50(8) (2000 Aug), pp. 1287-98.
- 492 Collins, J.; Brown, J.; Alexeeff, G.; Salmon, A. 1998. Potency Equivalency Factors for Some  
493 Polycyclic Aromatic Hydrocarbons and Polycyclic Aromatic Hydrocarbon Derivatives.  
494 *Regulatory Toxicology and Pharmacology*, 28 (1998, August), pp. 45-54. **DOI:**  
495 10.1006/rtph.1998.1235
- 496 Colman Lerner J.E., Orte M.A., Matamoros, N., Reyna Almandos J., Porta A. 2015. Estudio de  
497 hidrocarburos aromáticos policíclicos asociados al material particulado y en fase gaseosa en la  
498 ciudad de la plata y alrededores. En: *Contaminación atmosférica e hídrica en Argentina:*  
499 *Contribuciones del V Congreso PROIMCA y III Congreso PRODECA.* PULIAFITO E.; ALLENDE D.;  
500 PANIGATTI C., Eds. Universidad Tecnológica Nacional, 265 -274. ISBN 978-950-42-0163-2
- 501 Colman Lerner J.E.; Giuliani D.; Orte M.A.; Matamoros, N.; Sanchez, E.Y.; Porta A., 2016.  
502 Chapter 13-Volatile and semivolatile organic compounds determination in air. In: *The Quality*  
503 *of Air.* De La Guarda M., Ed. Elsevier, 73, (2016), pp. 321-342.  
504 <http://dx.doi.org/10.1016/bs.coac.2016.02.009>
- 505 Colman Lerner, J. E.; Morales, A.; Aguilar M.; Sánchez, E.; Giuliani, D.; Ditondo, J Massolo, L.;  
506 Doderio, V.; Porta, A. 2012. Concentración de compuestos orgánicos volátiles y material  
507 particulado en ambientes urbanos e industriales de dos regiones bonaerenses. Estudio  
508 Comparativo entre La Plata Bahía Blanca. *Actas 7mo Congreso de Medio Ambiente AUGM*, 22  
509 al 24 de mayo de 2012, UNLP, Argentina.  
510 [http://digital.cic.gba.gob.ar/bitstream/handle/11746/3457/11746\\_3457.pdf-PDFA--pdf-](http://digital.cic.gba.gob.ar/bitstream/handle/11746/3457/11746_3457.pdf-PDFA--pdf-PDFA.pdf?sequence=1&isAllowed=y)  
511 [PDFA.pdf?sequence=1&isAllowed=y](http://digital.cic.gba.gob.ar/bitstream/handle/11746/3457/11746_3457.pdf-PDFA--pdf-PDFA.pdf?sequence=1&isAllowed=y)
- 512 Dimashki, M.; Lim, L. H.; Harrison, R. H.; Harrad, S., 2001. Temporal Trends, Temperature  
513 Dependence, and Relative Reactivity of Atmospheric Polycyclic Aromatic Hydrocarbons. *Environ.*  
514 *Sci. Technol.*, 35, (2001, May), pp. 2264-2267. **DOI:** 10.1021/es000232y

- 515 Echeverri Londoño, C. A.; Maya Vasco, G. J. 2008. Relación entre partículas finas (PM 2,5) y  
 516 respirables (PM10) en la ciudad de Medellín. *Revista Ingenierías Universidad de Medellín*. Vol.7,  
 517 no. 12.
- 518 EU. 2008. European Parliament and of the Council Directive 2008/50/EC of the European  
 519 Parliament and of the Council of 21 May 2008 on ambient air quality and cleaner air for  
 520 Europe. *Official Journal of the European Union*, OJ L 152, 11.6.2008, pp. 1–44.
- 521 Fang, G.C., Huang, J.H., Huang, Y.L. 2010. Polycyclic aromatic hydrocarbons in the Asian  
 522 atmosphere during 2001–2009. *Environmental. Forensics*, 11 (2010, Sep) pp. 207–215. **DOI:**  
 523 10.1080/15275922.2010.495934
- 524 Fang, G.G; Wu, Y.S.; Chen, Jyh C; Chang, C.N.; Ho, T.T., 2006. Characteristic of polycyclic  
 525 aromatic hydrocarbon concentrations and source identification for fine and coarse particulates  
 526 at Taichung Harbor near Taiwan Strait during 2004-2005. *The Science of the Total Environment*  
 527 366, (2006), pp. 729-738. **DOI:** 10.1016/j.scitotenv.2005.09.075
- 528 Figueruelo, J.E.; Dávila, M.D. 2004. *Química física del ambiente y de los procesos*  
 529 *medioambientales*. Ed. Reverte.
- 530 Guevara, M. 2016. Emissions of Primary Particulate Matter. *Airborne Particulate Matter:*  
 531 *Sources, Atmospheric Processes and Health*, (2016), pp. 1-34. **DOI:** 10.1039/9781782626589-  
 532 00001
- 533 GutierrezJaimes J.H.; Quijano Vargas M.J.; Quijano Parra, A. 2012. Monitoreo y caracterización  
 534 fisicoquímica del material particulado PM<sub>2.5</sub> en Cúcuta-Norte de Santander-Colombia. *Revista*  
 535 *de la Facultad de Ciencias Basicas*, 10 (2012), pp.24-38.
- 536 <http://www.redalyc.org/pdf/903/90326398008.pdf>
- 537 IARC/WHO. 2013. World Health Organization. International Agency for Research on Cancer.  
 538 IARC: Outdoor air pollution a leading environmental cause of cancer deaths. Press Release Nº  
 539 221. [https://www.iarc.fr/en/media-centre/iarcnews/pdf/pr221\\_E.pdf](https://www.iarc.fr/en/media-centre/iarcnews/pdf/pr221_E.pdf)
- 540 INDEC. 2010. Instituto Nacional de Estadísticas y Censos. Cuadro P2-D. Provincia de Buenos  
 541 Aires, partido Tandil. Población total por sexo e índice de masculinidad, según edad en años  
 542 simples y grupos quinquenales de edad.
- 543 International Vehicle Emissions Model, 2008. Attachment D, Field data collection  
 544 activities. International Sustainable Systems Research Center - ISSRC.
- 545 IPCS. 2000. International Programme on Chemical Safety. Environment Health criteria 214;  
 546 Human Exposure Assessment. Geneva World Health Organization.
- 547 Jamhari, A. A.; Jamhari, M.; Sahani, M. T.; Latif, K. M.; Chan, H. S.; Tan, M. F.; Khan, N. M.;  
 548 Tahir, N. 2014. Concentration and source identification of polycyclic aromatic hydrocarbons  
 549 (PAHs) in PM10 of urban, industrial and semi-urban areas in Malaysia. *Atmospheric*  
 550 *Environment*, 86 (2014, Apr), pp. 16-27. **DOI:** 10.1016/j.atmosenv.2013.12.019.

- 551 Kong S, Li X, Li L, Yin Y, Chen K, Yuan L, Zhang Y, Shan Y, Ji Y. 2015. Variation of polycyclic  
552 aromatic hydrocarbons in atmospheric PM<sub>2.5</sub> during winter haze period around 2014 Chinese  
553 Spring Festival at Nanjing: Insights of source changes, air mass direction and firework particle  
554 injection. *Science of the Total Environment*, 520 (2015, Nov.), pp. 59-72. DOI:  
555 10.1016/j.scitotenv.2015.03.001
- 556 Kukkonen, J., Pohjola, M., Sokhi, R.S., Luhana, L., Kitwiroon, N., Fragkou, L., Rantamaki, M.,  
557 Berge, E., Ødegaard, V., Slørdal, L.H., Denby, B., Finardi, S. 2005. Analysis and evaluation of  
558 selected local-scale PM<sub>10</sub> air pollution episodes in four European cities: Helsinki, London,  
559 Milan and Oslo. *Atmospheric Environment* 39, (2005, May), pp. 2759-  
560 2773. <http://dx.doi.org/10.1016/j.atmosenv.2004.09.090>
- 561 Lewtas, J. 2007. Air pollution combustion emissions: characterization of causative agents and  
562 mechanisms associated with cancer, reproductive, and cardiovascular effects.  
563 *MutationResearch* 636 (2007, Nov), pp. 95-133. DOI: 10.1016/j.mrrev.2007.08.003
- 564 Marino, D. 2011. Estudio teórico experimental sobre respuestas biológicas a compuestos  
565 orgánicos de relevancia ambiental. Tesis doctoral. Facultad de Ciencias Exactas. Universidad  
566 Nacional de La Plata.
- 567 Markakis, K.; Poupkou, A.; Melas, D.; Zerefos, C. 2010. A GIS based anthropogenic PM<sub>10</sub>  
568 emission inventory for Greece. *Atmospheric Pollution Research* 1 (2010, April), pp. 71-81.  
569 <http://dx.doi.org/10.5094/APR.2010.010>
- 570 Massolo, L. 2004. Exposición a contaminantes atmosféricos y factores de riesgo asociados a la  
571 calidad de aire en La Plata y alrededores. Tesis de doctorado. UNLP. CIMA. 2004.
- 572 Massolo, L. Rehwagen, M.; Porta, A.; Ronco, A.; Herbarth, O.; Mueller, A. 2009. Indoor-  
573 Outdoor Distribution and Risk Assessment of Volatile Organic Compounds in the Atmosphere  
574 of Industrial and Urban Areas. *Environmental Toxicology*, 25 (2010, August), pp. 339-349. DOI:  
575 10.1002/tox.20504.
- 576 Massolo, L.; Müller, A.; Tueros, M.; Rehwagen, M.; Franck, U.; Ronco, A.; Herbarth, O. 2002.  
577 Assessment of Mutagenicity and Toxicity of Different-Size Fractions of Air Particulates from La  
578 Plata, Argentina, and Leipzig, Germany. *Environmental Toxicology*. 17 (2002, Jul), pp. 219-31.  
579 DOI: [10.1002/tox.10054](http://dx.doi.org/10.1002/tox.10054)
- 580 Mastandrea C., Chichizola C., Ludueña B., Sánchez H., Álvarez H., Gutiérrez A. 2005.  
581 Hidrocarburos aromáticos policíclicos. Riesgos para la salud y marcadores biológicos. *Acta*  
582 *BioquimClin* L, 39 (2005), pp. 27-36.
- 583 MSAL. 2012. Ministerio de Salud. Natalidad, mortalidad general, infantil y materna por lugar de  
584 residencia: Argentina-año 2010. Dirección de Estadísticas e Información de Salud. Boletín  
585 Número 134.
- 586 NEPC. 2006. National Environment Protection Council. National Environment Protection  
587 (national Pollutant Inventory) Measure. Technical Advisory Panel. Final Report to the National  
588 Environment Protection Council.



- 589 [http://www.scew.gov.au/sites/www.scew.gov.au/files/resources/759262ce-2dcd-0b74-6d3d-](http://www.scew.gov.au/sites/www.scew.gov.au/files/resources/759262ce-2dcd-0b74-6d3d-7605f819cd07/files/npi-rpt-technical-advisory-panel-final-report-final-200606.pdf)  
590 [7605f819cd07/files/npi-rpt-technical-advisory-panel-final-report-final-200606.pdf](http://www.scew.gov.au/sites/www.scew.gov.au/files/resources/759262ce-2dcd-0b74-6d3d-7605f819cd07/files/npi-rpt-technical-advisory-panel-final-report-final-200606.pdf)
- 591 NIOSH.1998. Polynuclear Aromatic Hydrocarbons by HPLC, Method 5506.NIOSH Manual of  
592 Analytical Methods (NMAM), Fourth Edition (1998, Jun), pp.1-9.
- 593 Nisbet, I.; LaGoy, P. 1992. Toxic Equivalency Factors (TEFs) for Polycyclic Aromatic  
594 Hydrocarbons (PAHs). Regulatory Toxicology and Pharmacology, 16 (1992), pp. 290-300.
- 595 NTP. 2016. National Toxicology Program. Report on Carcinogens, Fourteenth Edition; Research  
596 Triangle Park, NC: U.S. Department of Health and Human Services, Public Health Service.  
597 [http://www.niehs.nih.gov/health/materials/report\\_on\\_carcinogens\\_14th\\_edition\\_the\\_508.pdf](http://www.niehs.nih.gov/health/materials/report_on_carcinogens_14th_edition_the_508.pdf)  
598 f
- 599 OEHHA. 1994. Office of Environmental Health Hazard Assessment. OEHHA.Benzo(a)pyrene as a  
600 Toxic Air Contaminant. Part B. Health Assessment.Air Resources Board and Office of  
601 Environmental Health Hazard Assessment, California Environmental Protection Agency,  
602 Sacramento.
- 603 Ostro, B. 2004.World Health Organization. Outdoor air pollution: Assessing the environmental  
604 burden of disease at national and local levels. WHO Environmental Burden of Disease Series,  
605 No 5.
- 606 Perez L, Rapp R, Künzli N., 2010. The Year of the Lung: outdoor air pollution and lung health.  
607 Swiss Med Wkly, (2010, Nov.) 140:w13129, pp. 1-6. DOI: 10.4414/smw.2010.13129.
- 608 Perez L., Medina-Ramón M., Künzli N., Alastuey A., Pey J., Pérez N., Garcia R., Tobias A., Querol  
609 X., Sunyer J. 2009. Size fractionate particulate matter, vehicle traffic, and case-specific daily  
610 mortality in Barcelona, Spain. EnvironSciTechnol, 43(13) (2009 March), pp. 4707-4714.  
611 **DOI:** 10.1021/es8031488
- 612 Picone, N.; Campo, A. 2014. Comparación urbano-rural de parámetros meteorológicos de la  
613 ciudad de Tandil, Argentina. Revista de Climatología, 14 (2014), pp. 13-23.
- 614 Pufulete, M.; Battershill, J.; Boobis, A.; Fielder, R. 2004. Approaches to carcinogenic risk  
615 assessment for polycyclic aromatic hydrocarbons; a UK perspective. Regulatory Toxicology and  
616 Pharmacology, 40 (2004, August), pp. 54-66. <http://dx.doi.org/10.1016/j.yrtph.2004.04.007>
- 617 Schwartz J., Dockery D.W., Neas L.M. 1996. Is daily mortality associated specifically with fine  
618 particles? J Air WasteManagAssoc, 46(10) (1996, Oct), pp. 927-939.
- 619 Sosa, B.S.; Guerrero E.M; Banda Noriega, R.B. 2013a. Industria, Emisiones Gaseosas, y Gestión  
620 Ambiental Urbana. Diagnóstico de las condiciones operativas de las fundiciones en Tandil,  
621 Argentina. Revista Gestión y Ambiente, 16 (2013, May), pp. 61-70.
- 622 Sosa, B.S.; Banda Noriega, R.B.; Guerrero E.M. 2013b. Industrias de Fundición: aspectos  
623 ambientales e indicadores de condición ambiental. Revista de Metalurgia, 49, (2013, Jan-Feb),  
624 pp. 5-19.

- Sosa, B.S. 2015. Chapter 3- Marco metodológico; Chapter 5-Ambiente urbano: actividades antrópicas y fuentes contaminantes. En: Contaminación ambiental por material particulado y compuestos orgánicos volátiles en la ciudad de Tandil, provincia de Buenos Aires. Doctor en Ciencias Exactas, área Química; Facultad de Ciencias Exactas, Universidad Nacional de La Plata. Pp. 68-73 and pp. 104-105.
- The World Bank. 2014. World Development Indicators: Traffic and congestion. Charper 3.13.Disponible en <http://wdi.worldbank.org/table/3.13#>
- USEPA.1986. United States Environment Protection Agency.Health effect assessment for polycyclic aromatic hydrocarbon (PAH).EPA 540/1 – 86 – 013.Environmental Criteria an Assessment Office. Cincinnati, OH.
- <https://nepis.epa.gov/Exe/ZyNET.exe/2000FD6E.txt?ZyActionD=ZyDocument&Client=EPA&Index=1986%20Thru%201990&Docs=&Query=&Time=&EndTime=&SearchMethod=1&TocRestrict=n&Toc=&TocEntry=&QField=&QFieldYear=&QFieldMonth=&QFieldDay=&UseQField=&IntQFieldOp=0&ExtQFieldOp=0&XmlQuery=&File=D%3A%5CZYFILES%5CINDEX%20DATA%5C86THRU90%5CTXT%5C00000007%5C2000FD6E.txt&User=ANONYMOUS&Password=anonymous&SortMethod=h%7C-&MaximumDocuments=1&FuzzyDegree=0&ImageQuality=r75g8/r75g8/x150y150g16/i425&Display=hpfr&DefSeekPage=x&SearchBack=ZyActionL&Back=ZyActionS&BackDesc=Results%20page&MaximumPages=1&ZyEntry=1>
- USEPA.2009a. Environmental Protection Agency.Revised Appendix B\_RAGS Tables. Table 6.2 Cancer Toxicity Data -- Inhalation Wastebeds 1 - 8: Bike Trail GEDDES. New York, pp:26. 2009/01/23.<http://www.epa.gov/region02/water/lakes/Onondaga%20Bike%20Trail%20HHRA%20Appendix%20B.pdf>
- USEPA.2009b. US Environmental Protection Agency. Final Report: Integrated Science Assessment for particulate matter. EPA / 600 / R-08 / 139F. Washington, DC. [https://cfpub.epa.gov/si/si\\_public\\_record\\_report.cfm?dirEntryId=216546](https://cfpub.epa.gov/si/si_public_record_report.cfm?dirEntryId=216546)
- USEPA. 2013. Environmental Protection Agency, National Ambient Air Quality Standards for Particulate Matter, Final Rule. 3086 Federal Register, Rules and Regulations, 78 January 15, 2013; <http://www.gpo.gov/fdsys/pkg/FR-2013-01-15/pdf/2012-30946.pdf>
- Vargas, M.; Romero, G.; Palencia, A.; Rivero, E.; Piñero, S. 2013. Toxic equivalency factors of polycyclic aromatic hydrocarbons (PAHS) in Valencia, Venezuela. ActaToxicológicaArgenina, 21 (2013), pp. 69-77.
- Vardoulakis, S., Kassomenos, P. 2008. Sources and factors affecting PM10 levels in two European cities: Implications for local air quality management. Atmospheric Environment 42, (2008), pp. 3949-3963.
- Vicente AB., T. Sanfeliu, MM. Jordan. 2012. Assesment of PM10 pollution episodes in a ceramic cluster (NE Spain): Proposal of a new quality index for PM10, As, Cd, Ni and Pb. Journal of Environmental Management 108 (2012), pp. 92-101.

- WHO. 2000. World Health Organization. Air quality guidelines for Europe, Vol 91, 2 edn., World Health Organization, Copenhagen.  
[http://www.euro.who.int/\\_\\_data/assets/pdf\\_file/0005/74732/E71922.pdf](http://www.euro.who.int/__data/assets/pdf_file/0005/74732/E71922.pdf)
- WHO. 2006. World Health Organization. WHO Air quality guidelines, for particulate matter, ozone, nitrogen dioxide and sulfur dioxide. Global update 2005. Summary of risk assessment. [http://apps.who.int/iris/bitstream/10665/69477/1/WHO\\_SDE\\_PHE\\_OEH\\_06.02\\_eng.pdf](http://apps.who.int/iris/bitstream/10665/69477/1/WHO_SDE_PHE_OEH_06.02_eng.pdf)
- WHO. 2011. World Health Organization. Urban outdoor air pollution database. Department of Public Health and Environment, World Health Organization, Geneva, Switzerland.
- WHO. 2012. World Health Organization. IARC: diesel engine exhaust carcinogenic. [http://www.iarc.fr/en/media-centre/pr/2012/pdfs/pr213\\_E.pdf](http://www.iarc.fr/en/media-centre/pr/2012/pdfs/pr213_E.pdf)
- WHO. 2013. World Health Organization, Regional Office for Europe. Health effects of particulate matter: Policy implications for countries in eastern Europe, Caucasus and central Asia. [http://www.euro.who.int/\\_\\_data/assets/pdf\\_file/0006/189051/Health-effects-of-particulate-matter-final-Eng.pdf](http://www.euro.who.int/__data/assets/pdf_file/0006/189051/Health-effects-of-particulate-matter-final-Eng.pdf)
- WHO. 2014. World Health Organization. Burden of disease from the joint effects of Household and Ambient Air Pollution for 2012.  
[http://www.who.int/phe/health\\_topics/outdoorair/databases/AP\\_jointeffect\\_BoD\\_results\\_March2014.pdf?ua=1](http://www.who.int/phe/health_topics/outdoorair/databases/AP_jointeffect_BoD_results_March2014.pdf?ua=1)
- WHO. 2016. World Health Organization. WHO Global Urban Ambient Air Pollution Database (update 2016). Ambient (outdoor) air pollution database, by country and city. [http://www.who.int/phe/health\\_topics/outdoorair/databases/cities/en/](http://www.who.int/phe/health_topics/outdoorair/databases/cities/en/)
- Wiriya, W.; Prapamontol, T.; Chantara, S. 2013. PM<sub>10</sub>-bound polycyclic aromatic hydrocarbons in Chiang Mai (Thailand): Seasonal variations, source identification, health risk assessment and their relationship to air-mass movement. *Atmospheric Research*, 124 (2013, April), pp. 109-122. DOI: 10.1016/j.atmosres.2012.12.014.
- Zanobetti A, Schwartz J. 2009. The effect of fine and coarse particulate air pollution on mortality: a national analysis. *Environ Health Perspect*, 117(6) (2009 Jun), pp. 898-903. DOI: 10.1289/ehp.0800108
- Yang, H.H.; Lee, W.J.; Chen, S.J.; Lai, S.O. 1998. PAH emission from various industrial stacks. *Journal of Hazardous Materials*, 60 (1998, June), pp. 159-174. [http://dx.doi.org/10.1016/S0304-3894\(98\)00089-2](http://dx.doi.org/10.1016/S0304-3894(98)00089-2)

**Highlights**

Urban PM and PAHs from point and mobile sources represent a health risk for citizens.

Pollutant characterisation according to land use areas facilitates the study of the sources involved and provides useful data for management.

Health risk due to PM exposure is a state indicator (diagnosis) of ambient air which allows to evaluate the relationship between anthropogenic activities and the deterioration of air quality